$5 - x \geq 0 \Rightarrow 5 \geq x$
$\Rightarrow x \leq 5$

SECTION A [Multiple choice questions] - 12 marks

QUESTION 1

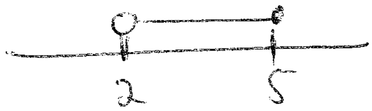
The largest interval on which the function $f(x) = \dfrac{\sqrt{5-x}}{\sqrt{x-2}}$ is continuous, is

and

$x - 2 > 0 \Rightarrow x > 2$

(a) ϕ	(b) $[2,\infty)$	(c) $(-\infty, 5]$	(d) $(2,5)$	(e) $(2,5]$	(f) $[2,5]$	(g) None of (a) - (f)

QUESTION 2



Consider the function $F(x) = f(xf(x))$ and the table below.

$F(x) = f(xf(x))$
$F'(x) = f'(xf(x)) \times \frac{d}{dx}(x f(x))$
$= f'(xf(x)) \times [f(x) + xf'(x)]$

x	2	3	4	6
$f(x)$	3	4	4	5
$f'(x)$	4	1	8	3

$F'(2) =$ $F'(2) = f'(2f(2)) \times [f(2) + 2f'(2)] = f'(6)(3 + 2(4))$

(a) 3	(b) 5	(c) 12	(d) 15	(e) 21	(f) 33	(g) None of (a) - (f)

$= 3(11)$
$= 33.$

QUESTION 3

$\lim\limits_{x\to 3^+} \ln(x-3)^{-1} =$ $-\lim\limits_{x \to 3^+} \ln(x-3)$

$x \to 3^+$
$\Rightarrow x > 3$

(a) ∞	(b) $-\infty$	(c) 0	(d) $\frac{1}{3}$	(e) $-\frac{1}{3}$	(f) None of (a) - (e)

$\Rightarrow x - 3 > 0$
$\Rightarrow -\lim\limits_{x\to 3^+} \ln(x-3) = -(-\infty)$
$= \infty.$

QUESTION 4

Consider the following statements:

(1) $\sinh(t) + \cosh(t) = e^t$; ✓

(2) $\lim\limits_{t\to\infty} \sinh(t) = \infty$; ✓

(3) $\dfrac{d^2}{dt^2}\sinh(t) = -\sinh(t)$. ✗

Choose the correct answer:

(a) Only statement (1) is correct.

(b) Only statement (2) is correct.

(c) Only statement (3) is correct.

(d) Only statements (1) and (2) are correct.

(e) Only statements (1) and (3) are correct.

(f) Only statements (2) and (3) are correct.

(g) Statements (1), (2) and (3) are correct.

(h) Statements (1), (2) and (3) are false.

(1) $\sinh t + \cosh t = \dfrac{e^t + e^{-t}}{2} + \dfrac{e^t - e^{-t}}{2}$
$= e^t$

(2) $\lim\limits_{t\to\infty} \sinh(t) = \lim\limits_{t\to\infty} \dfrac{e^t + e^{-t}}{2}$
$= \infty$

(3) $\dfrac{d}{dt}(\sinh t) = \cosh t$
$\dfrac{d^2}{dt^2}(\sinh t) = \dfrac{d}{dt}(\cosh t)$
$= \sinh t.$

QUESTION 5

Consider the following statements:

(1) If $\lim_{x\to 1} f(x) = \infty$ then the line $x = 1$ is a vertical asymptote of x.

(2) If f is continuous at a, then f is differentiable at a;

(3) $\lim_{x\to 1} \dfrac{x^2+6x-7}{x^2+5x-6} = \dfrac{\lim_{x\to 1}(x^2+6x-7)}{\lim_{x\to 1}(x^2+5x-6)}$;

Choose the correct answer:

(a) Only statement (1) is correct.

(b) Only statement (2) is correct.

(c) Only statement (3) is correct.

(d) Only statements (1) and (2) are correct.

(e) Only statements (1) and (3) are correct.

(f) Only statements (2) and (3) are correct.

(g) Statements (1), (2) and (3) are correct.

(h) Statements (1), (2) and (3) are false.

QUESTION 6

Consider the following statements:

(1) If $f(x) = 3\ln(3x)$, then $f'(x) = \dfrac{3}{x}$.

(2) If $f(x) = \sin(3x) + \cos(3x)$, then $f''(x) + 9f(x) = 0$.

(3) If $f(x) = \sin(x)$, then $f'(x)^2 + f(x)^2 = 1$.

Choose the correct answer:

(a) Only statement (1) is correct.

(b) Only statement (2) is correct.

(c) Only statement (3) is correct.

(d) Only statements (1) and (2) are correct.

(e) Only statements (1) and (3) are correct.

(f) Only statements (2) and (3) are correct.

(g) Statements (1), (2) and (3) are correct.

(h) Statements (1), (2) and (3) are false.

$\frac{d}{dx}(3\ln 3x)$

$= 3 \times \frac{1}{3x} \times 3 = \frac{3}{x}$.

$f'(x) = 3\cos 3x - 3\sin 3x$

$f''(x) = -9\sin 3x - 9\cos 3x$

$= -9(\sin 3x + \cos 3x)$

$f''(x) + 9f(x)$

$= -9(\sin 3x + \cos 3x) + 9(\sin 3x + \cos 3x)$

$= 0$

$f(x) = \sin x$

$f'(x) = \cos x$

$(f'(x))^2 = \cos^2 x$

$\lim_{x\to 0} \frac{\sin x}{e^{6x}-1} \quad \left(\frac{0}{0}\right)$

L'Hosp. $= \lim_{x\to 0} \frac{\cos x}{6e^{6x}} = \frac{1}{6}$

QUESTION 7

$\lim_{x\to 0} \frac{\sin(x)}{e^{6x}-1} =$ $\left(\frac{0}{0}\right)$

(a) $\frac{1}{6}$	(b) $-\frac{1}{6}$	(c) 0	(d) 1	(e) $\frac{1}{5}$	(f) ∞	(g) None of (a) - (f)

QUESTION 8

Consider $f(x) = \arctan(x) + 4^x$. Then $f''(x) =$

(a) $\frac{1}{1+x^2} + 4^x(\ln 4)$	(b) $\frac{-2x}{(1+x^2)^2} + 4^x(\ln 4)$	(c) $\arctan(x)\sec^2(x) + 4^x(\ln 4)^2$	(d) $\frac{2x}{(1+x^2)^2} + 4^x(\ln 4)^2$
(e) $\arctan(x)\sec^2(x) + 4^x$	(f) $\frac{-2x}{(1+x^2)^2} + 4^x(\ln 4)^2$	(g) $\arctan(x)\sec^2(x) + 4^x(\ln 16)$	(h) None of these.

End of Section A

ROUGH WORKING

$f'(x) = \left(\frac{1}{1+x^2}\right) + 4^x \ln 4 = (1+x^2)^{-1} + 4^x \ln 4$

$f''(x) = (-1)(1+x^2)^{-2} \times 2x + 4^x(\ln 4)^2$

$= \frac{-2x}{(1+x^2)^2} + 4^x(\ln 4)^2$

SECTION B (13 marks): Write only your final answer in the block provided. Only this will be marked. Rough working can be done outside the blocks and will not be marked.

QUESTION 9 [2 marks]

Consider $y = \dfrac{1+\sin(x)}{\cos(x)}$.

If $x = \pi$: $y = \dfrac{1+\sin\pi}{\cos\pi} = \dfrac{1}{-1} = -1$

(9.1) Write down the derivative of y.

$$\frac{dy}{dx} = \frac{\cos x(\cos x) - (1+\sin x)(-\sin x)}{\cos^2 x} = \frac{1+\sin x}{\cos^2 x}$$

(9.2) Write down the equation of the line that is tangent to the curve of y where $x = \pi$.

$$y = x - 1 - \pi.$$

$$\left.\frac{dy}{dx}\right|_{x=\pi} = \frac{1+\sin\pi}{\cos^2\pi} = \frac{1}{1} = \underline{1} = m$$

$y = mx + c = 1x + c$

$-1 = \pi + c \Rightarrow$

$\underline{c = -1-\pi}$

QUESTION 10 [1 mark]

Do not simplify your answer. $\dfrac{d}{dx}\cos(\ln(x)) =$

$$-\sin(\ln(x)) \times \frac{1}{x}$$

QUESTION 11 [1 mark]

Consider the function $f(x) = \begin{cases} cx^2 + 2x & \text{if } x < 2 \\ x^3 - cx & \text{if } x \geq 2 \end{cases}$

The value(s) of the constant c for which the function f is continuous on the real line, is(are):

$$c = 2/3$$

$\lim_{x \to 2^-} cx^2 + 2x = 4c + 4$

$\lim_{x \to 2^+} x^3 - cx = 8 - 2c$

$8 - 2c = 4c + 4$

$\Longrightarrow c = 2/3$

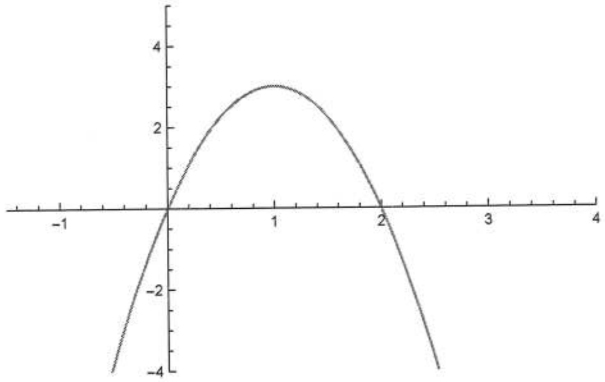
QUESTION 12 [1 mark]

Complete the following part of the Concavity Test:
The graph of a function f is concave upward on an interval I if

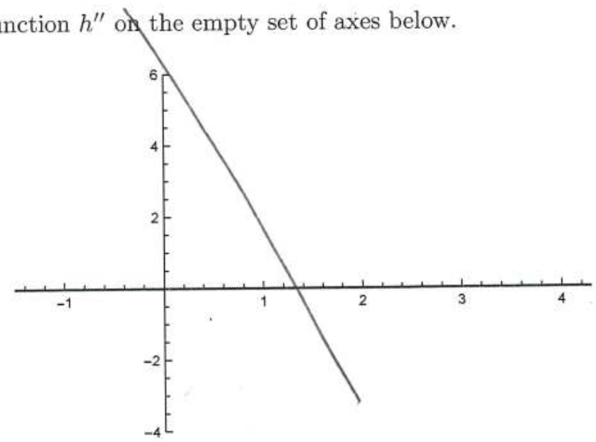
$f''(x) > 0$ for all $x \in I$.

QUESTION 13 [4 marks]

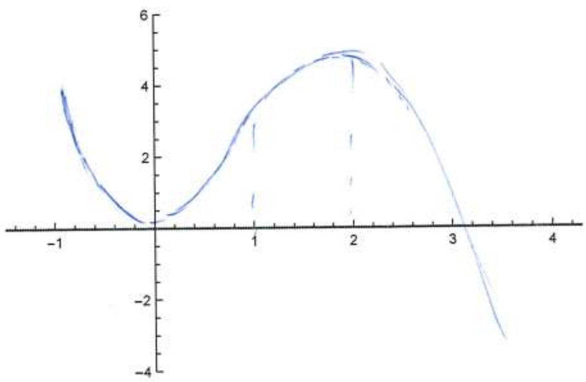
Consider the graph of h', defined on the real line, given below.



(a) Draw the graph of the function h'' on the empty set of axes below. (1)



(b) Draw the graph of the function h on the empty set of axes below, given that $h(0) = 0$. (1)



(c) The inflection point(s) on the curve of h is(are) at $x =$ (1)

$\boxed{1}$

(d) The graph of h is concave upward on the interval: (1)

$\boxed{(-\infty, 1)}$

Only this

Also correct: $(-\infty, 1]$.

QUESTION 14 [2 marks]

The value(s) of c that satisfies (satisfy) the Mean Value Theorem for the function $f(x) = \dfrac{1}{x}$ on the interval $[1, 4]$, is(are) given by:

$\boxed{c = 2.}$

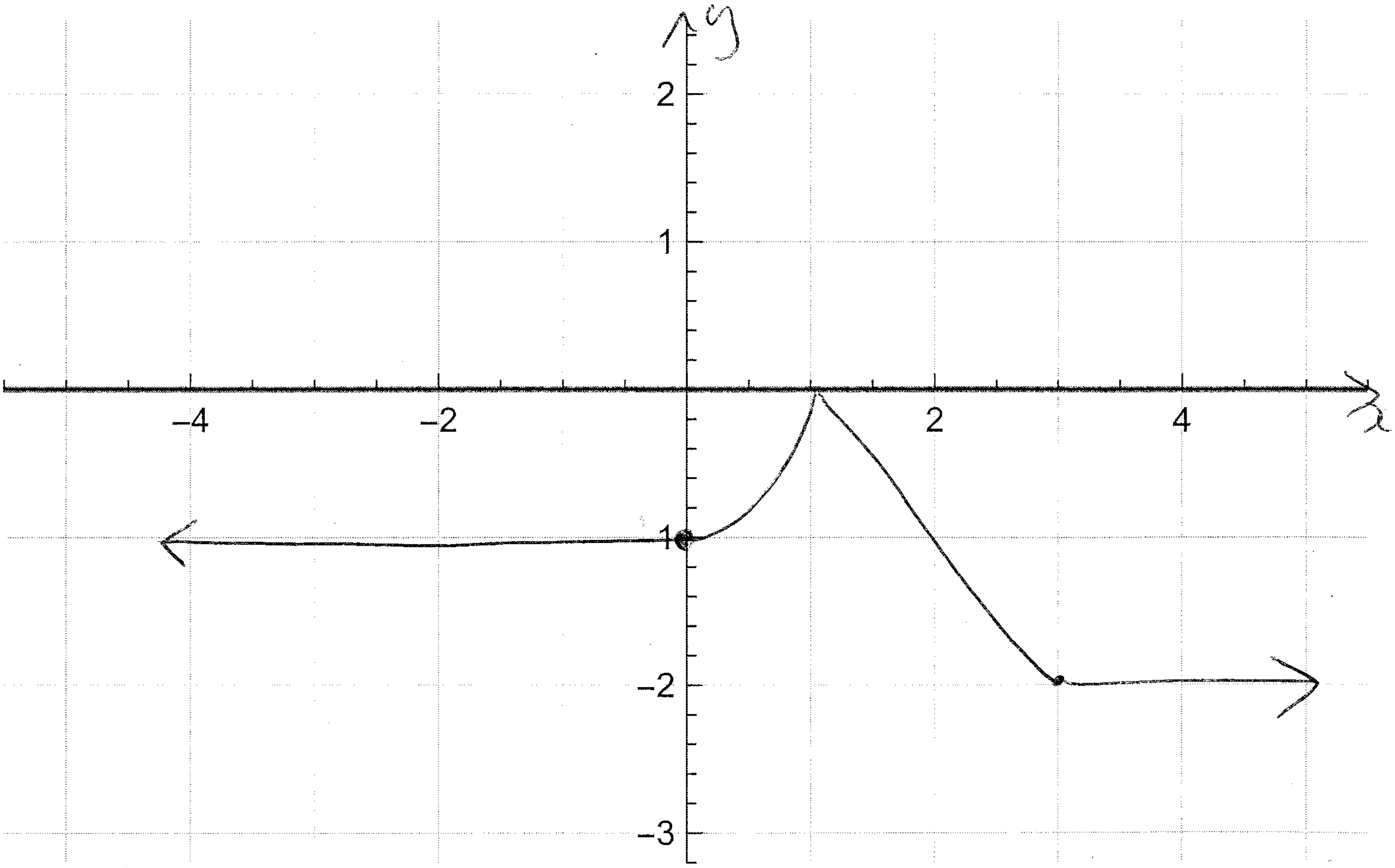
$f'(x) = \dfrac{-1}{x^2}$

$f'(c) = \dfrac{f(4) - f(1)}{4 - 1} = \dfrac{\frac{1}{4} - 1}{4 - 1} = \dfrac{\frac{1-4}{4}}{4-1} = \dfrac{-1}{4} = \dfrac{-1}{c^2}$

QUESTION 15 [2 marks]

Sketch the graph of the function f on the axes provided, indicating all relevant intercepts and information, where f is continuous on $\mathbb{R}$ and

$$f(0) = -1, f'(x) = 0 \text{ if } x < 0 \text{ and } x \geq 3, f'(x) = 2x \text{ if } 0 < x < 1, f'(x) = -1 \text{ if } 1 < x < 3.$$



End of Section B

SECTION C [Full written answers] - 20 marks

QUESTION 16 [3 marks]

Use the Closed Interval Method to determine the extreme values of $f(t) = (t^2 - 4)^3$ on the interval $[-2, 3]$.

$$f'(t) = 3(t^2-4) \times 2t$$
$$= 6t(t-2)^2(t+2)^2$$

$f'(t) = 0 \Rightarrow t = -2,\ t = 0$ or $t = 2$

$\therefore$ Critical points: $t = -2, 0, 2$.

$f(-2) = 0$
$f(0) = -64$
$f(2) = 0$
$f(3) = 125$

Abs max is $f(3) = 125$

Absolute minimum is $f(0) = -64$.

QUESTION 17 [5 marks]

(a) Show that $\lim_{x\to 0} \frac{\cos(x)-1}{x} = 0$. Do not use L'Hospital's Rule. (3)

$$LHS = \lim_{x\to 0} \frac{\cos x - 1}{x} \quad \left(\frac{0}{0}\right)$$

$$= \lim_{x\to 0} \left(\frac{\cos x - 1}{x}\right)\left(\frac{\cos x + 1}{\cos x + 1}\right)$$

$$= \lim_{x\to 0} \frac{\cos^2 x - 1}{x(\cos x + 1)} \quad \left(\frac{0}{0}\right)$$

$$\sin^2 x + \cos^2 x = 1$$
$$\cos^2 x - 1 = -\sin^2 x$$

$$= \lim_{x\to 0} \frac{-\sin^2 x}{x(\cos x + 1)}$$

$$= \lim_{x\to 0} \frac{-\sin x}{x} \times \lim_{x\to 0} \frac{\sin x}{\cos x + 1}$$

$$= -\lim_{x\to 0} \frac{\sin x}{x} \times \lim_{x\to 0} \frac{\sin x}{\cos x + 1}$$

$$= -1 \times 0$$

$$= 0.$$

$$= RHS$$

(b) Find the limit or show that it does not exist: $\lim_{x\to\infty} \ln(1+x) - \ln(2+x)$. (2)

$$\lim_{x\to\infty} \ln(1+x) - \ln(2+x) \quad (\infty - \infty)$$

$$= \lim_{x\to\infty} \ln \frac{1+x}{2+x}$$

$$= \ln \lim_{x\to\infty} \frac{1+x}{2+x}$$

$$= \ln 1$$

$$= 0$$

$$\lim_{x\to\infty} \frac{1+x}{2+x} \quad \left(\frac{\infty}{\infty}\right)$$

$$= \lim_{x\to\infty} \frac{x(\frac{1}{x}+1)}{x(\frac{2}{x}+1)}$$

$$= \lim_{x\to\infty} \frac{1+\frac{1}{x}}{1+\frac{2}{x}}$$

$$= 1$$

OR: $\lim_{x\to\infty} \frac{1+x}{2+x} \quad \left(\frac{\infty}{\infty}\right)$

$$\stackrel{\text{L'Hospital}}{=} \lim_{x\to\infty} \frac{1}{1}$$

$$= 1$$

QUESTION 19 [6 marks]

(a) Use implicit differentiation to find $\frac{dy}{dx}$ if $y\sec(x) = \tan(xy)$. (3)

$$\frac{d}{dx}\left(y\sec x\right) = \frac{d}{dx}\tan(xy)$$

$$\Rightarrow \left(\frac{dy}{dx}\right)\sec x + y(\sec x\tan x) = \sec^2(xy) \times \frac{d}{dx}(xy)$$

$$\Rightarrow \frac{dy}{dx}\sec x + y(\sec x\tan x) = \sec^2(xy)\left(y + x\frac{dy}{dx}\right)$$

$$= (\sec^2 xy)y + x\sec^2(xy)\frac{dy}{dx}$$

$$\Rightarrow \frac{dy}{dx}\left(\sec x - x\sec^2(xy)\right) = y\sec^2(xy) - y\sec x\tan x$$

$$\Rightarrow \frac{dy}{dx} = \frac{y\sec^2(xy) - y\sec x\tan x}{\sec x - x\sec^2(xy)}.$$

(b) Use logarithmic differentiation to find $\frac{dy}{dx}$ if $y = \frac{(x+1)^5 \ln(x)}{e^{2x}\sqrt{2x-3}}$. (3)

$$\ln y = \ln\left(\frac{(x+1)^5 \ln x}{e^{2x}\sqrt{2x-3}}\right)$$

$$= \ln(x+1)^5 + \ln(\ln x) - \ln e^{2x} - \ln(2x+3)^{\frac{1}{2}}$$

$$= 5\ln(x+1) + \ln(\ln x) - 2x - \frac{1}{2}\ln(2x+3)$$

$$\Rightarrow \frac{1}{y}\frac{dy}{dx} = \frac{d}{dx}\left(5\ln(x+1) + \ln(\ln x) - 2x - \frac{1}{2}\ln(2x+3)\right)$$

$$= \frac{5}{x+1} + \frac{1}{\ln x} \times \frac{1}{x} - 2 - \frac{1}{2}\left(\frac{1}{2x+3}\right) \times 2$$

$$\Rightarrow \frac{dy}{dx} = \frac{(x+1)^5 \ln x}{e^{2x}\sqrt{2x-3}}\left(\frac{5}{x+1} + \frac{1}{x\ln x} - 2 - \frac{1}{2x+3}\right)$$

y.

QUESTION 20 [3 marks]

Find all horizontal and vertical asymptotes of $y = \dfrac{2e^x}{e^x - 5}$.

Horizontal asymptotes:

Find $\lim_{x\to\infty} \frac{2e^x}{e^x-5}$ and $\lim_{x\to-\infty} \frac{2e^x}{e^x-5}$

$\lim_{x\to\infty} \frac{2e^x}{e^x-5} \quad \left(\frac{\infty}{\infty}\right)$

$= \lim_{x\to\infty} \frac{2e^x}{e^x} \quad$ (L'Hospital)

$= 2$

$\lim_{x\to-\infty} \frac{2e^x}{e^x-5} \quad \left(\frac{0}{-5}\right)$

$= 0$

$\Rightarrow$ $y=0$ and $y=2$ are Horizontal asymptotes.

Vertical asymptotes:

We can have a vertical asymptote where

$e^x - 5 = 0 \Rightarrow e^x = 5 \Rightarrow x = \ln 5$

$\lim_{x\to(\ln 5)^-} \frac{2e^x}{e^x-5} \quad \left(\frac{2}{\text{small neg}}\right)$

$= -\infty$

$x \to (\ln 5)^-$

$\Rightarrow x < \ln 5$

~~$\Rightarrow x - \ln 5 < 0$~~

$\Rightarrow e^x < 5$

$\Rightarrow e^x - 5 < 0$

$\boxed{\text{Can also find } \lim_{x\to(\ln 5)^+} \frac{2e^x}{e^x-5}}$

$\Rightarrow x = \ln 5$ is a vertical asymptote.

QUESTION 21 [3 marks]

Prove the following theorem: If f is differentiable at a, then f is continuous at a.

Given: f is differentiable at a

$$\Rightarrow f'(a) = \lim_{x \to a} \frac{f(x) - f(a)}{x - a}.$$

Required To prove: f continuous at a

$$\Rightarrow \lim_{x \to a} f(x) = f(a)$$

Proof

$$\lim_{x \to a} f(x) = \lim_{x \to a} (f(x) - f(a) + f(a))$$

$$= \lim_{x \to a} (f(x) - f(a)) + \lim_{x \to a} f(a)$$

$$= \lim_{x \to a} \frac{(f(x) - f(a))(x - a)}{x - a} + f(a)$$

$$= \lim_{x \to a} \frac{f(x) - f(a)}{x - a} \times \lim_{x \to a} (x - a) + f(a)$$

$$= f'(a) \times 0 + f(a)$$

$$= f(a)$$

$\Rightarrow f$ is continuous at a.

End of Section C